Huawei Access Node HEB-ACC-33 QoS Congestion Event Resolution SOP

Node: HEB-ACC-33 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_3549

Description

This SOP details the procedure for resolving QoS congestion events on Huawei Access Node HEB-ACC-33 at the Hebbal Edge site. This failure domain is primarily attributed to Microbursts and interface tail drops due to the inherent limitations of QoS queue management within the Huawei Access Node. The QoS algorithm, in an attempt to prioritize high-priority packets, can lead to a cascading effect of queue starvation, resulting in congestion events. Furthermore, the Huawei Access Node's hardware architecture and software configuration can exacerbate this issue, leading to the occurrence of ALARM_3549. It is essential to understand the underlying protocol mechanics, including the QoS scheduling algorithm and its interaction with the Access Node's hardware, to effectively troubleshoot and resolve this issue.

Pre-requisite Checks

- check: Confirm the site location as Hebbal Edge and the affected node as HEB-ACC-33
- check: Verify the redundant link status to ensure high availability
- check: Dispatch remote hands to Hebbal Edge for assistance if required

Potential Root Causes

Cause	Probability
QoS queue starvation due to high-priority packet prioritization	High
Microbursts and interface tail drops due to hardware limitations	Medium
Insufficient QoS configuration or misconfiguration	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue status, Expected Output: Queue status: normal, Queue depth: 0 packets, Description: This command is used to check the current status of the QoS queue, including its depth and any potential issues

Command: display interface brief, Expected Output: Interface status: up, Interface type: 10GE, Description: This command is used to verify the status of the interface and its type, helping to identify potential hardware-related issues

Command: display qos schedule, Expected Output: QoS scheduling: enabled, Schedule type: WFQ, Description: This command is used to check the QoS scheduling configuration, including the enabled status and schedule type, which can help identify potential
```

```
configuration-related issues
```

Resolution Steps

Step 1: Step: Modify QoS configuration to improve queue management, Command: qos queue modify queue 1 weight 50, Description: This command modifies the QoS configuration to improve queue management by adjusting the weight of queue 1 to 50, which can help alleviate queue starvation

Step 2: Step: Adjust QoS scheduling to reduce prioritization of high-priority packets, Command: qos schedule modify schedule 1 priority 10, Description: This command adjusts the QoS scheduling to reduce the prioritization of high-priority packets, which can help mitigate the occurrence of congestion events

Step 3: Step: Verify and adjust QoS configuration as necessary, Command: display qos queue status, Description: This command is used to verify the status of the QoS queue and adjust the configuration as necessary to ensure optimal performance

Post-Check Verification

```
Command: display qos queue status, Expected Output: Queue status: normal, Queue depth: 0 packets, Description: This command is used to verify the status of the QoS queue after making changes to the configuration

Command: display interface brief, Expected Output: Interface status: up, Interface type: 10GE, Description: This command is used to verify the status of the interface after making changes to the configuration

Command: display qos schedule, Expected Output: QoS scheduling: enabled, Schedule type: WFQ, Description: This command is used to verify the QoS scheduling configuration after making changes to the configuration
```

Roll-back Procedures

- Step: Restore original QoS configuration, Command: qos queue restore, Description: This command restores the original QoS configuration, which can help mitigate the occurrence of congestion events

- Step: Revert QoS scheduling changes, Command: qos schedule restore, Description: This command reverts the QoS scheduling changes made during the troubleshooting process